

PLANET FITNESS MEMBER RETENTION & ENGAGEMENT ANALYSIS

Data-Driven Insights with Innovative Concepts for Growth

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PROJECT OVERVIEW

This data analytics project explores **member retention and gym usage behavior** at Planet Fitness **using a multi-year synthetic dataset**. The goal was to simulate real-world operational challenges faced by a fitness chain and apply **data-driven insights to improve member engagement**, reduce churn, and optimize resources.

Key components of the project include **exploratory analysis, predictive modeling, and behavioral segmentation**.



DATA & METHODOLOGY

Dataset Summary

- Synthetic dataset created to simulate real-world behavior across a Planet Fitness location
- Covers Jan 2022 – Dec 2024
- Includes:
 - 1,500+ members
 - 100,000+ gym check-ins
 - Member details: age, gender, join/cancel date, ZIP code, membership type
 - Timestamped check-in data (date + time of day)



DATA & METHODOLOGY

Tools & Technologies

- Python (Pandas, NumPy) for data processing
- Matplotlib & Seaborn for visualizations
- scikit-learn for churn prediction modeling
- Jupyter Notebook for analysis and iterative development



DATA & METHODOLOGY

Analytical Methods

- **Retention analysis:** Time-based bins to study membership duration trends
- **Check-in heatmaps:** Hour-by-day breakdown of gym traffic
- **Churn prediction model:** Random Forest classifier based on early activity behavior
- **Behavior segmentation:** Comparison by membership type and activity level
- **Scenario modeling:** What-if analysis of a performance-based pricing strategy



KEY FINDINGS

Member Retention Insights

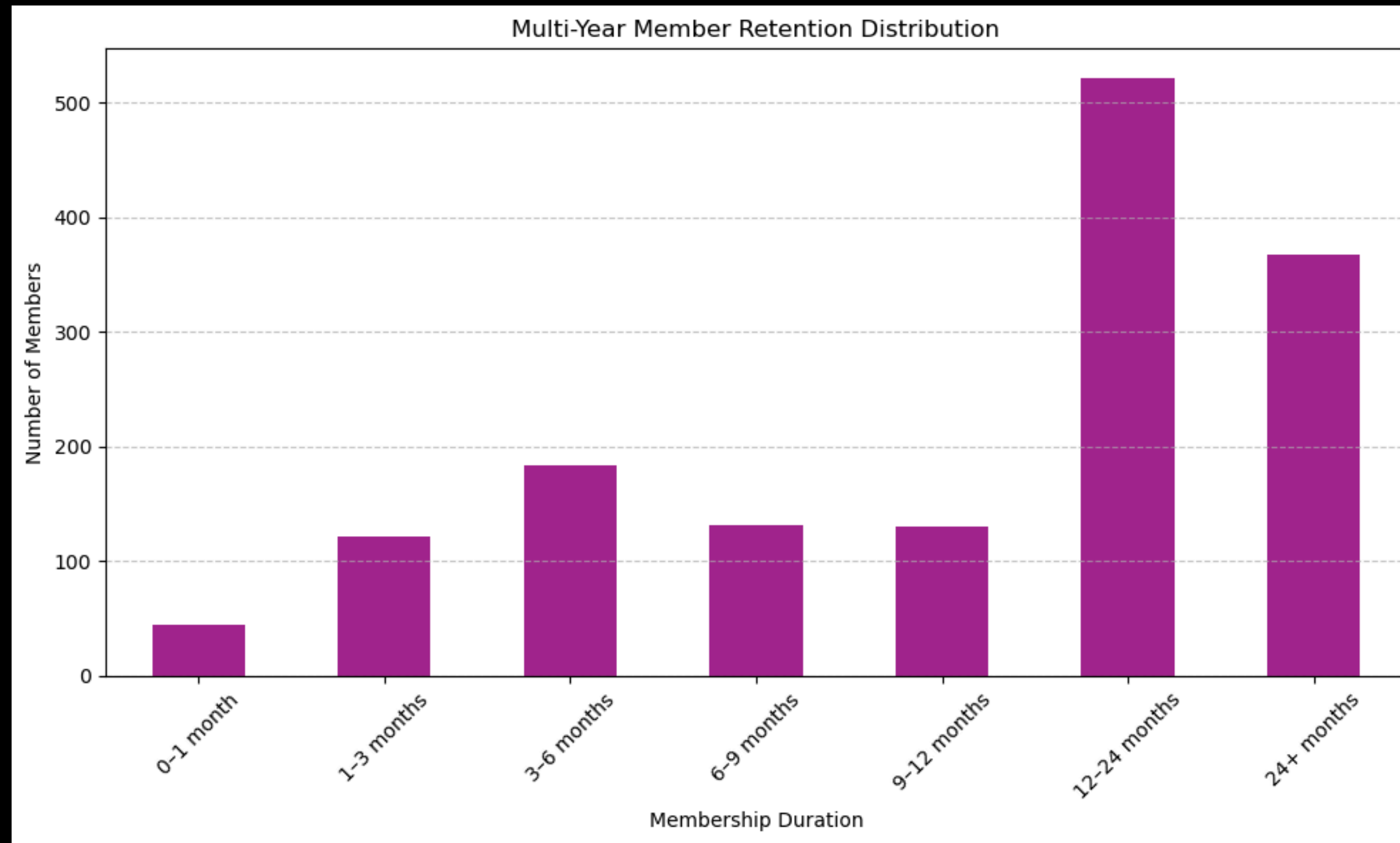
- **Over 35% of members churn** within the first 3 months
- Members who check in frequently during their first 30 days are **significantly more likely to stay**
- Longer-term members tend to have **more consistent weekly check-ins**

Usage Behavior Patterns

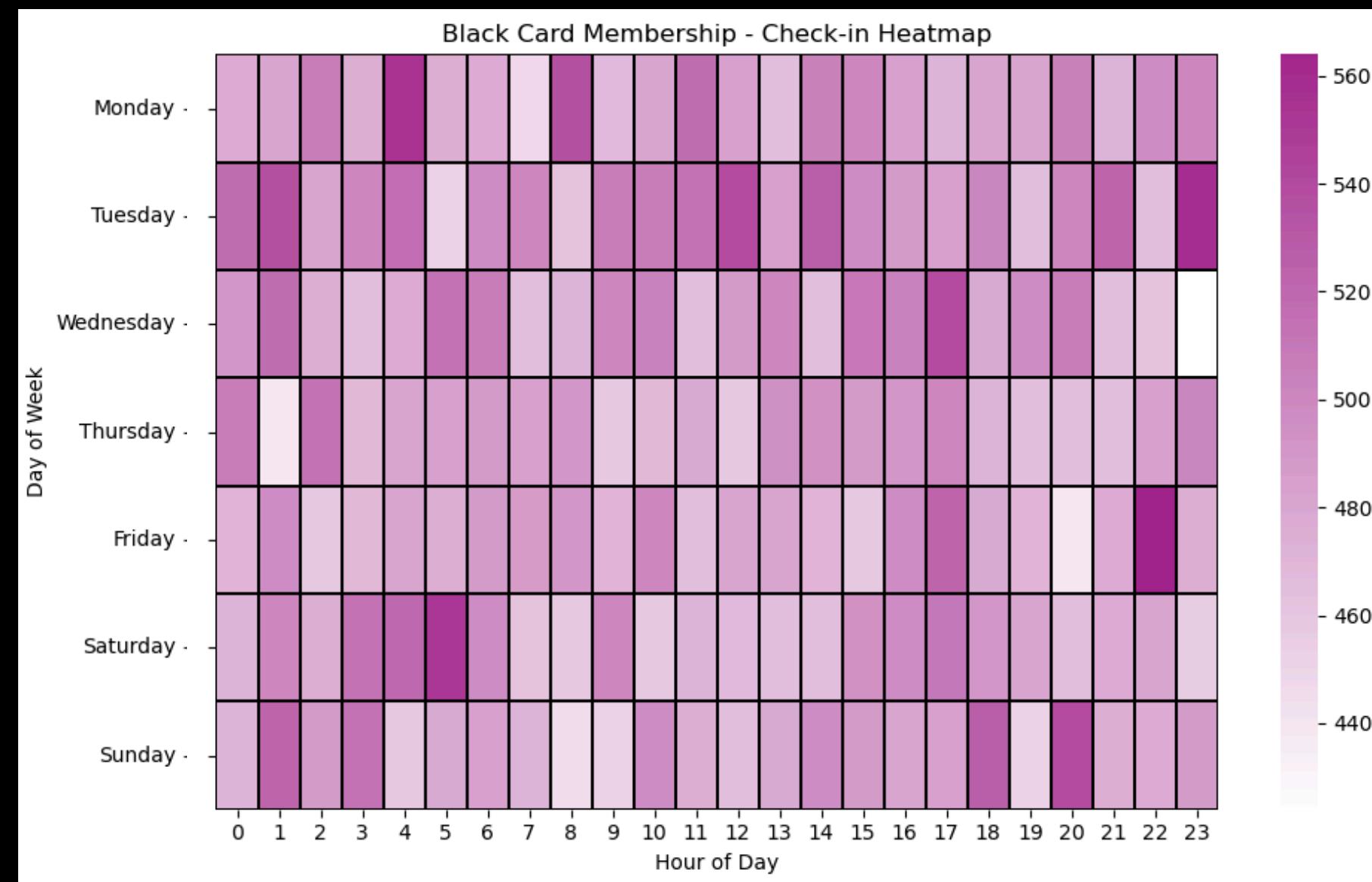
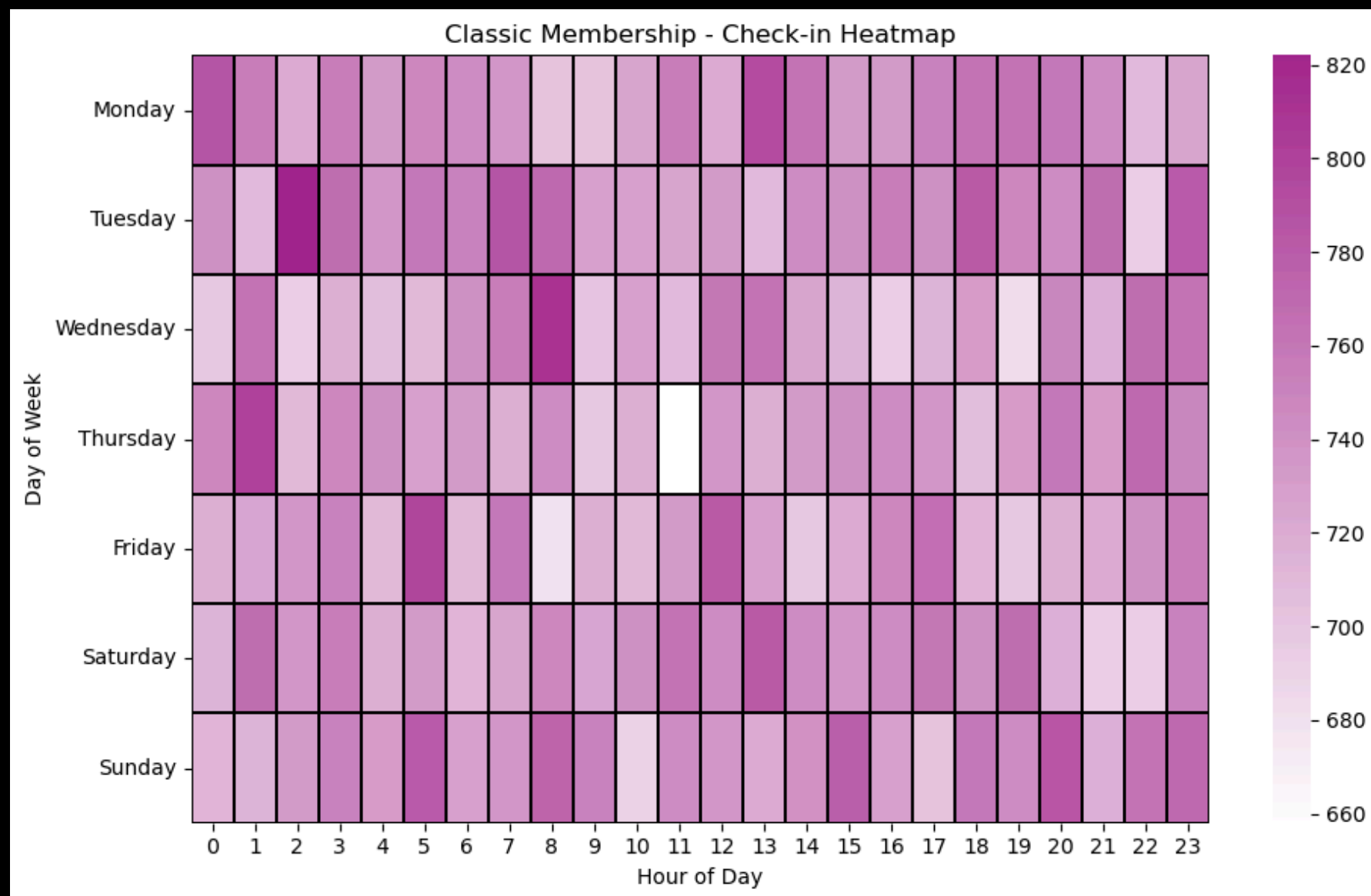
- Check-in activity peaks between **6–9 AM and 5–8 PM**
- Monday and Tuesday are the **most active days**; Saturday late night is **least active**
- Black Card members check in **more frequently** and across **more varied hours** than Classic members



VISUALIZATIONS

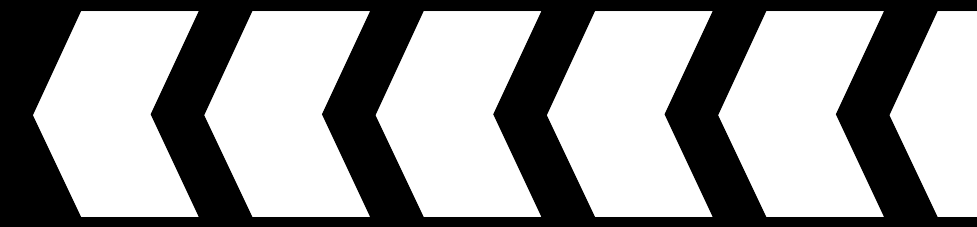


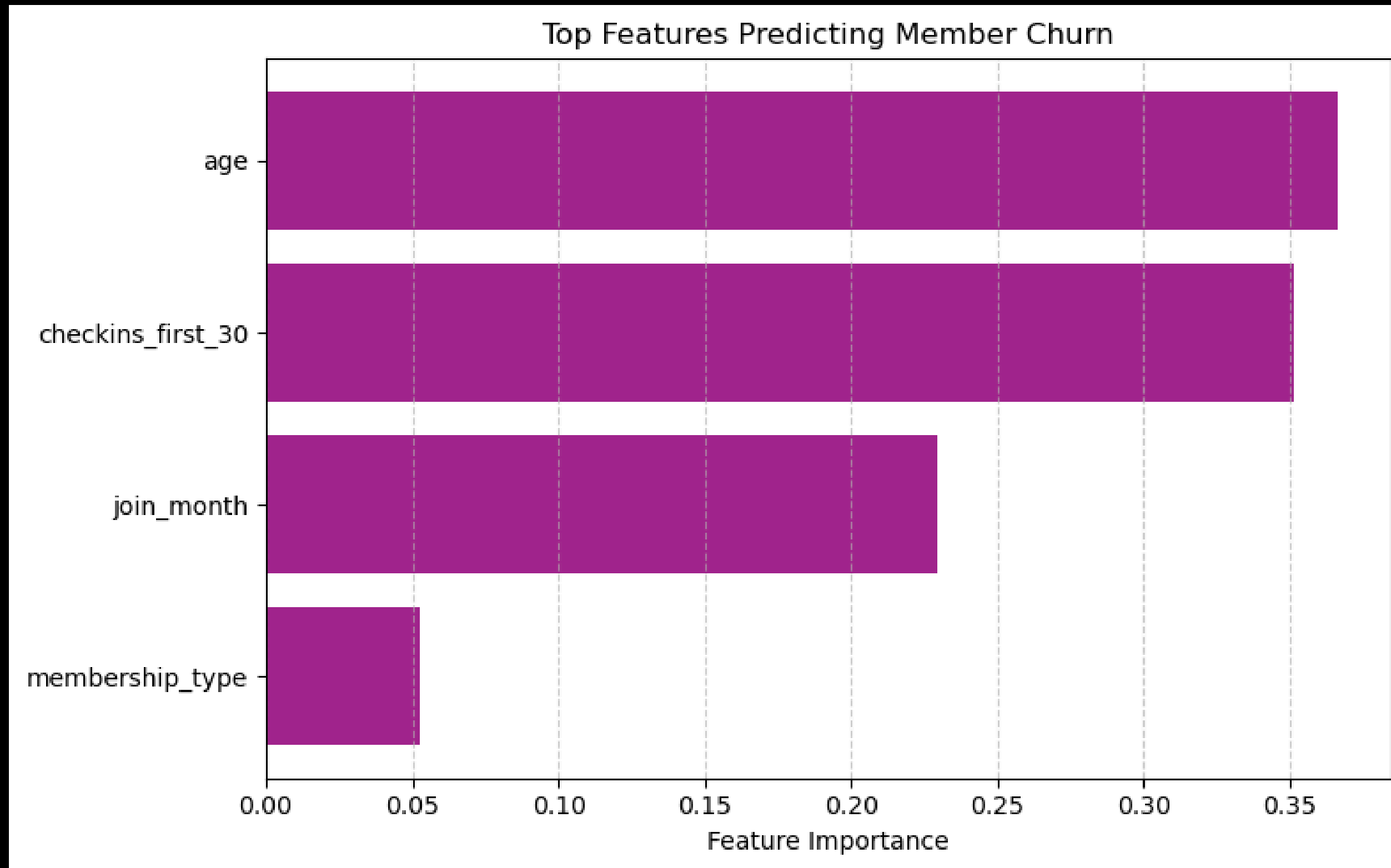
Over 35% of members churn within the first 3 months



Check-in heatmaps: Hour-by-day breakdown of gym traffic

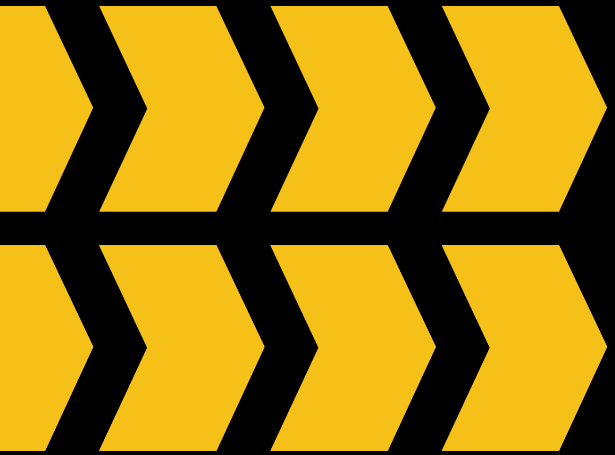
Black Card members check in more frequently and across more varied hours than Classic members





Members who check in frequently during their first 30 days are significantly more likely to stay





STRATEGIC RECOMMENDATIONS

Target At-Risk Members Early

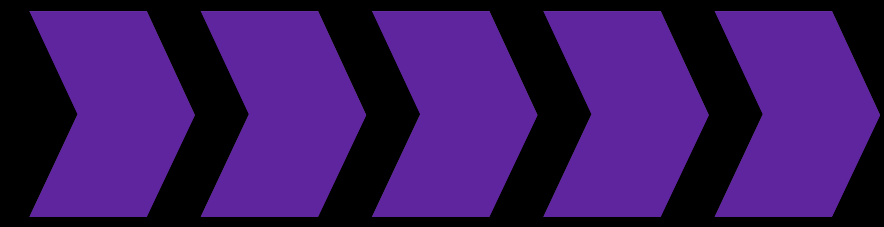
- Use churn prediction models to identify members with low check-in frequency in their first 30 days. Target them with automated follow-ups, check-in challenges, or support.

Optimize Staffing by Usage Patterns

- Heatmap analysis reveals consistent daily and hourly trends (e.g., 6–9 AM and 5–8 PM peaks). Use this data to optimize staffing, cleaning schedules, and maintenance planning.

Segment Strategies by Membership Type

- Classic and Black Card members show different usage behaviors. Tailor retention campaigns and incentives based on usage frequency and time-of-day preferences.



CONTACT

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The image is a composite of two photographs. On the left, a woman with brown hair tied back, wearing a black sports bra and leggings, is smiling while holding a large black weight plate. She is in a gym setting with metal racks visible in the background. On the right, a woman with dark hair in a bun, wearing a black sports bra, is performing a burpee on a paved surface. A large black diagonal bar runs across the center of the image, serving as a background for the text.

THANK YOU

[HTTPS://NICOLEGALLO.GITHUB.IO/NICKELCOIN/](https://nicolegallo.github.io/nickelcoin/)